

Blockchain Experiment 4

Name: Atharva Lotankar

Class: D20C **R.No.:** 20

Date: 04 / 09 / 2026

Aim: Creation of account in Metamask, Transfer of Amount/Cryptocurrency from one account to another.

Theory:

1. MetaMask Overview

MetaMask is a cryptocurrency wallet used to create and manage blockchain accounts. It allows users to send, receive, and manage cryptocurrencies and interact with blockchain applications.

2. Account Creation

A MetaMask account consists of:

- Public Address – Used to receive cryptocurrency.
- Private Key – Used to authorize transactions.
- Secret Recovery Phrase – Used to recover the wallet.

The recovery phrase and private key must be kept secret and secure.

3. Wallet Address

Every MetaMask account has a unique wallet address. This address is shared with others when receiving cryptocurrency. Blockchain transactions are recorded using the sender's and receiver's addresses.

4. Cryptocurrency Transfer

Cryptocurrency can be transferred from one MetaMask account to another by entering the receiver's wallet address and the amount to be transferred. The transaction is then reviewed and confirmed by the sender.

5. Transaction Confirmation

After confirmation, the transaction is broadcast to the blockchain network. Once verified and included in a block, the transfer is recorded on the blockchain. A unique Transaction Hash (Tx Hash) is generated to track the transaction.

6. Gas Fee

Blockchain transactions require a network fee, commonly called a gas fee on Ethereum-based

networks. The sender must have sufficient funds to pay this fee in addition to the transfer amount.

7. Security

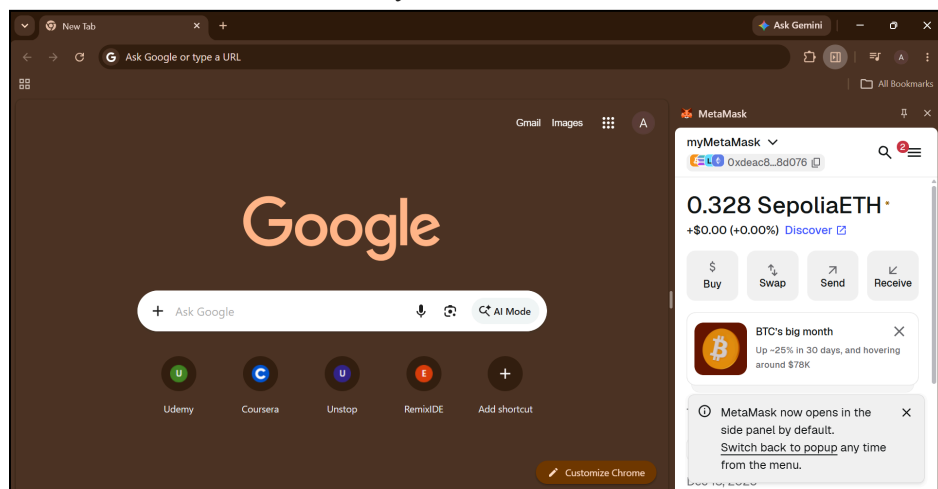
Blockchain transactions are generally irreversible. Therefore, the receiver's address must be verified before sending cryptocurrency. The private key and Secret Recovery Phrase should never be shared.

8. Tools Used

- MetaMask – For creating accounts and transferring cryptocurrency.
- Blockchain Network – For processing and recording transactions.
- Web Browser – For accessing MetaMask.
- Blockchain Explorer – For verifying transactions using the transaction hash.

Procedure & Output:

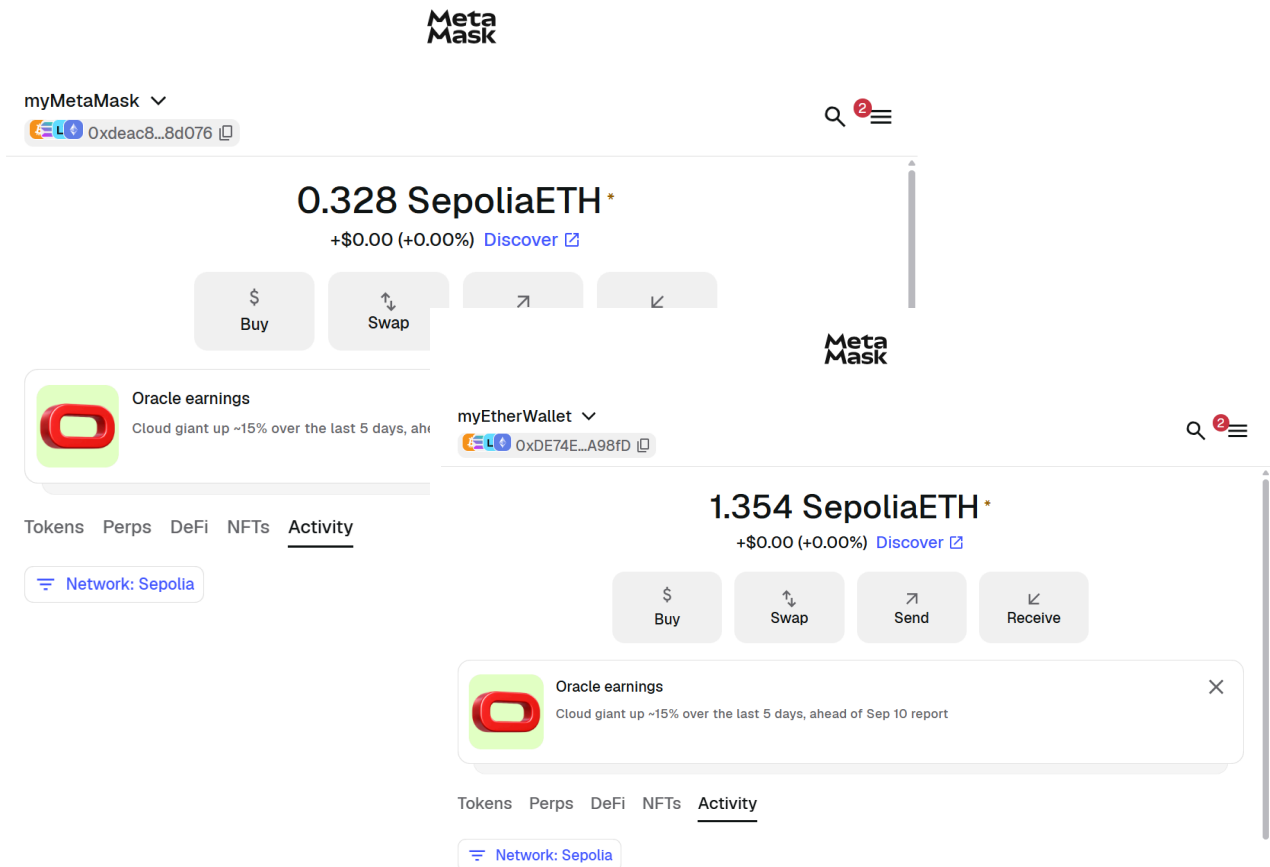
1. Install the MetaMask extension in the web browser.
2. Open MetaMask and select Create a New Wallet.
3. Set a password and securely save the Secret Recovery Phrase.
4. Complete the wallet setup to create the account.
5. Note/copy the public wallet address of Account 1.
6. Create or select another MetaMask account (Account 2).
7. Copy the wallet address of Account 2.
8. Switch back to Account 1 and select Send.
9. Enter Account 2's wallet address and the amount of cryptocurrency to transfer.
10. Review the transaction amount and gas fee.
11. Confirm the transaction.
12. Wait for the transaction to be processed by the blockchain.
13. Verify that the amount has been received in Account 2.
14. Check the Transaction Hash to verify the transaction details.



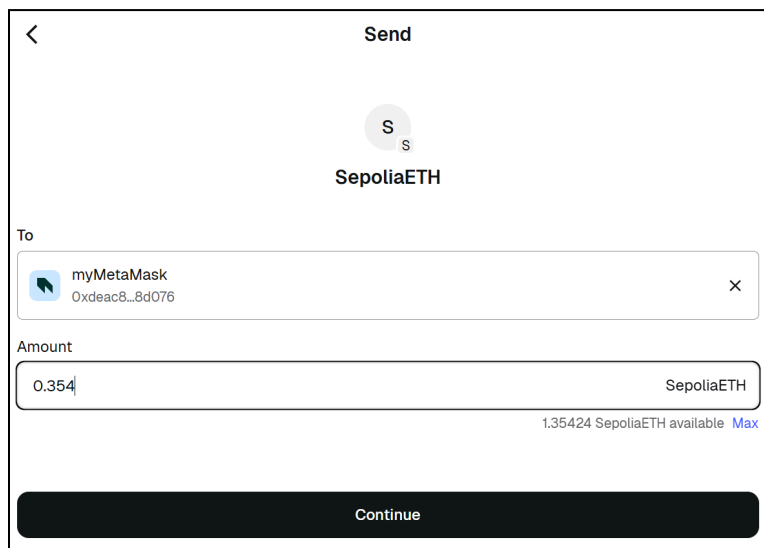
Home Screen: MetaMask Wallet

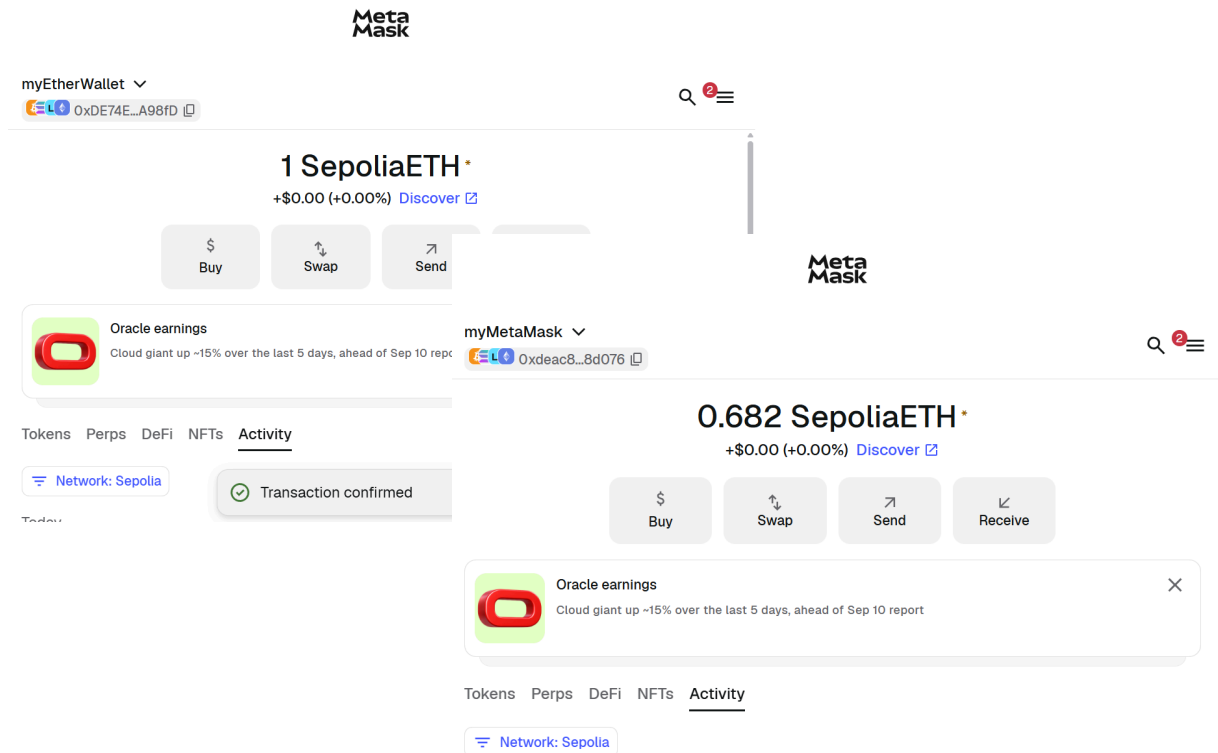
Sending Specific SepEth from 'myEtherWallet' Account to 'myMetaMask' Account

Before Balance:

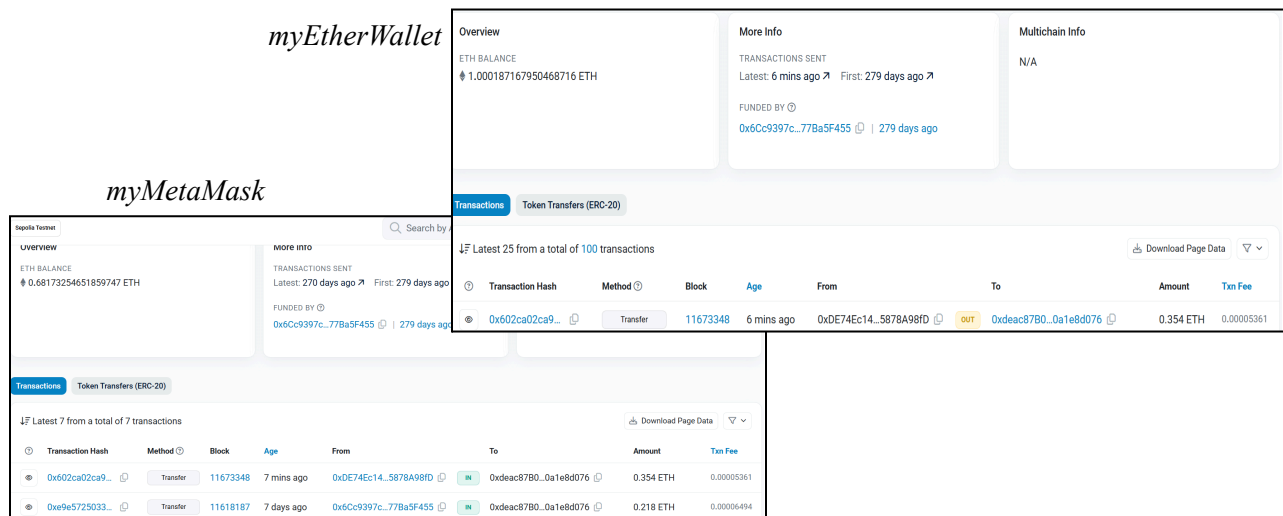


After Transaction:





Transaction Passbook: <https://sepolia.etherscan.io/>



Conclusion: The experiment successfully demonstrated the creation and management of MetaMask accounts and the transfer of cryptocurrency between two accounts. The transaction was confirmed using the wallet address and transaction hash, providing practical understanding of blockchain transactions, gas fees, and wallet security.